

CS Capstone Design

Technical Demo Grading Sheet (100 pts)

TEAM: Heartline

Overview: The main purpose of the “Technical Demos” is to very clearly communicate the extent to which the team has identified key challenges in the project, and has proven solutions to those challenges. Grading is based on how complete/accurate the list of challenges is, and how convincingly and completely the given demos cover the given challenges.

This template is fleshed out by the team, approved by CS mentor, and brought to demo as a grading sheet.

Risky technical challenges

Based on our requirements acquisition work and current understanding of the problem and envisioned solution, the following are the key technical challenges that we will need to overcome in implementing our solution:

C1. Twine Game reading/writing from/to an external Database: This aspect of the project is of utmost importance to our client, as the point of the project is to gather research data for Clinicians to better understand and assist Indigenous communities. The challenge is enabling Twine to connect with an external Database, and then make the calls necessary to make operations over it.

C2. Individual User Tracking: This is an expectation of the Lab, the ability for a user to individually track their healthcare journey and see their growth over time. The challenge in development is the storing of related data. For any given user, they should only have access to data relating to their account.

C3. Communal Goal Reaching: This is a less-major goal of the Lab, the ability for any user to see their individual effort shared with their community in the name of reaching a goal unachievable by any single person. This challenge is largely one of data management, the Back-end ability to gather data, clean it for use, sum up the totals, and deliver it to the user in a timely manner.

C4. Navigation within Flutter: This is a secondary aspect of our project, as members of the Lab have been designing our shell for months now. However, within our section of the App, we are still expected to create a coherent shell for hosting the individual Twine Games. The Challenge is allowing a user to navigate through our section.

C5. Importing and Running Twine within Flutter: This aspect of the project is largely expected to be completed by our Team Lead, Pavan Gorintla, however, it is important to prove it's something we are capable of. For this challenge, we need to see a Flutter App hosting a Twine Game.

Challenges covered by demos:

In this section, we outline the demonstrations we have prepared, and exactly which of the challenge(s) each one of them proves a solution to.

Demonstration 1: Zombie Apocalypse Game

Challenges addressed:

C1. Twine Game Reading/Writing from/to an external Database

C2. Individual User Tracking

C3. Communal Goal Reaching

Flight Plan:

1. Twine game allows for a user to write their name to the database.
2. Database produces an ID that is stored in the database alongside their name, and returns the ID to the user. The user then uses their name and given ID to log into the database (**C1.**)
3. Once logged in, users are met with a dashboard that allows them to self-report their exercise activities to the database. (**C2.**)
4. In the same dashboard, users can view a list of all previously submitted exercise activities, and allow them to delete submissions at will. (**C2.**)
5. In another passage of the Twine Game, users can view their contributions tallied up with the contributions of every other user. (**C3.**)

Evaluation:

✓ Convincingly demo'd each of listed challenges?

✓ Other evaluative comments:

Demonstration 2: Flutter Hosting a Twine Game

Challenges addressed:

C4. Navigation within Flutter

C5. Importing and Running Twine within Flutter

Flight Plan:

1. A basic Flutter app that has two buttons. Each button navigates to a different page within the app, the user can then return to this main page (**C4**).
2. Each of the sub-pages of the app hosts a very basic “Hello World” style Twine Game (**C5**).

Evaluation:

✓ Convincingly demo'd each of listed challenges?

✓ Other evaluative comments:

Other challenges recognized by not addressed by demo:

If there were challenges you listed earlier that were *not* covered by a demo, list here. This will hopefully be a short list...but better to be clear about where you are. If you have items here, you could list (if applicable) any pending plans to reduce these risks.